Photosynthesis 14a

1 Read the text below and use the following words to complete the passage:

green pigment	are combined	because	Chlorophyll	most plants
the sun	they	This energy		

All energy needed for life to exist on Earth comes from _____ . _____ is trapped by green plants in a process called photosynthesis. During photosynthesis carbon dioxide and water _____ to make glucose (a kind of sugar) and oxygen. For plants to photosynthesise _____ also need a _____ called chlorophyll. _____ is responsible for absorbing light in all producers. This is the reason why _____ are green. Plants are also called autotrophs _____ they make their own food.

2 Define the term 'producer'. _____

3 Complete the word equation for photosynthesis below.

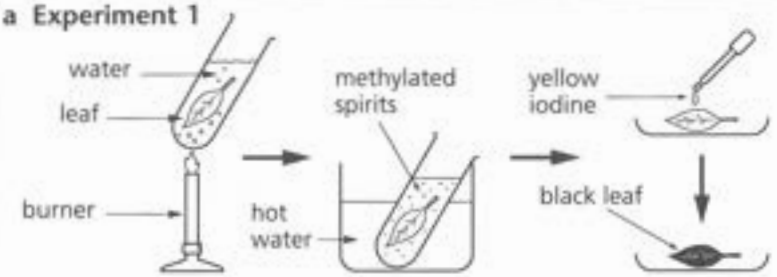
b _____

a _____ + water ⟶ sugar + d _____

c _____

4 A science class performed the following experiments to test various aspects of the word equation. Carefully read through each procedure and write what you think the aim and conclusion would be.

a Experiment 1



- A plant leaf was boiled and placed in a test tube of methylated spirits, which was then placed in very hot water.
- When the chlorophyll was removed, the leaf was rinsed and placed on a petri dish.
- Iodine solution was added to the leaf. It turned from yellow to blue/ black. Starch was present.

Aim:

_____

_____

_____

Conclusion:

_____

_____

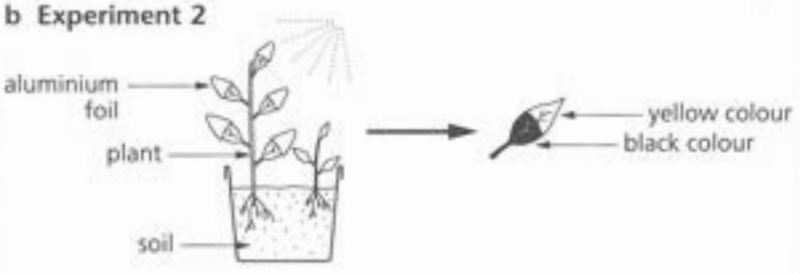
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b Experiment 2



- Using aluminium foil, the top half of individual leaves on a plant were covered.
- The plant was then placed under fluorescent light for 3 days.
- A leaf was then removed and tested for starch using the same steps as in experiment 1.

Aim:

_____

_____

_____

Conclusion:

_____

_____

_____

_____

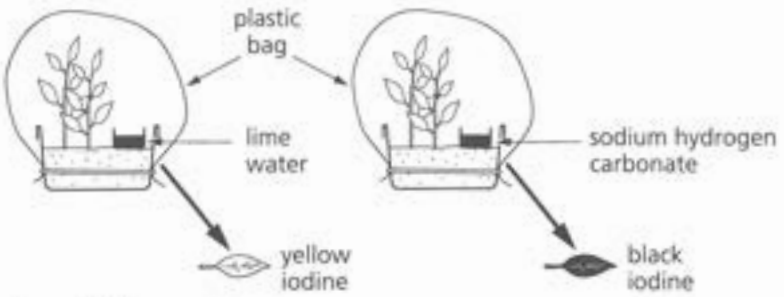
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Photosynthesis 14b

c Experiment 3

(*Note:* Limewater absorbs CO_2 out of air; sodium hydrogen carbonate releases CO_2 into the air.)

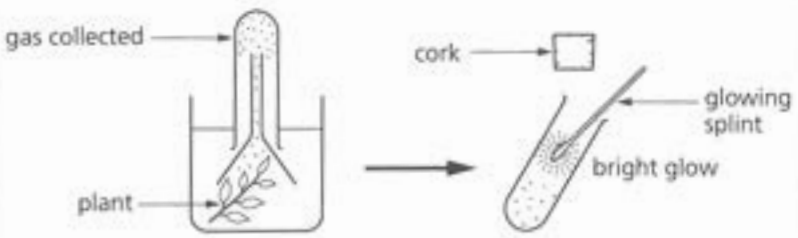


- A petri dish containing limewater was placed into a potted plant.
- A second dish containing sodium hydrogen carbonate was placed into a second pot.
- Plants were left under fluorescent light for 3 days and tested for starch using iodine.

Aim:

Conclusion:

d Experiment 4



- A plant was exposed to fluorescent light for 4 days.
- The resulting gas was collected in a test tube, which was then corked and tested with a glowing splint.
- Splint glowed brighter.

Aim:

Conclusion:

5 Write up one of the experiments above as an experimental record.

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